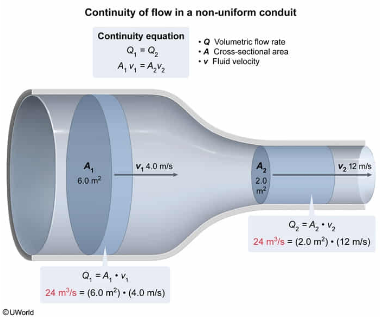
An ideal fluid flows through a tube at an initial speed of 4.0 m/s. What will be the speed of the fluid if the cross-sectional area of the tube decreases from 6.0 m^2 to 2.0 m^2?

- A. 1.3 m/s [7%]
- B. 4.0 m/s [2%]
- ✔ C. 12 m/s [81%]
- D. 36 m/s [8%]

Correct
82% answered correctly

Explanation:



The term *flow* refers to the quantity of a fluid (ie, liquid or gas) that passes a point per unit of time. Flow may take place in open spaces or within a conduit (a pipe or channel through which a fluid passes). The volume of flow that takes place within an enclosed conduit (often expressed in units of m^3/s) is given by the volumetric flow rate (Q), which is the product of the cross-sectional area (A) of the conduit and the velocity of the fluid (v):

$$Q = A \cdot v$$

Calculating the volumetric flow rate may be useful in assessing flow through both uniform and non-uniform conduits. In fact, the volumetric flow rate through interconnected segments of *any* conduit must be equal (Q_1 present in the first segment must equal Q_2 present in a second adjoining segment). This relationship is expressed as the **continuity equation**, shown below:

$$Q_1 = Q_2$$

This consistency of volumetric flow rate between segments can be explained by the characteristics of ideal fluids, which cannot be compressed in accordance with container geometry. As a result, inter-segmental differences in a cross-sectional area (A) in the conduit necessitate changes in the velocity of fluid flow (v). Accordingly, the continuity equation can also be expressed as:

$$A_1 \cdot v_1 = A_2 \cdot v_2$$

In this scenario, an ideal fluid is initially flowing with a velocity of 4.0 m/s through a segment of an enclosed conduit with a cross-sectional area of 6 m^2. The same fluid then flows into the next segment of the tube with a cross sectional area of 2 m^2. Inputting this information into the second form of the continuity equation allows for the calculation of the fluid velocity within the second conduit segment:

$$A_1 \cdot v_1 = A_2 \cdot v_2$$

$$(6.0\ \text{m}^2) \cdot (4.0$$

$$\frac{24\ \text{m}^3/\text{s}}{2.0\ \cancel{\text{m}}^2} = v_2$$

$$v_2 = 12\ \text{m}/\text{s}$$

Accordingly, the **velocity of the fluid** will increase three-fold from 4.0 m/s to 12 m/s **(Choice C)** as it reaches the conduit segment with a cross-sectional area that is one-third the size of the initial conduit segment.

Educational objective:
The volumetric flow rate of ideal fluids through interconnected segments of any conduit must be equal at every point along the length of the conduit. The continuity equation can be used to relate volumetric flow rate of one conduit segment to another.

Time spent: 0 sec
Subject: Physics

QID: 402039
System: Fluids